



Open Source SystemVerilog Parsing and Linting

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<https://github.com/AsFigo/SVALint>

Outline

- Introduction
- SVALint architecture
- Examples with SVALint

Introduction

Open Source Extensible Linting

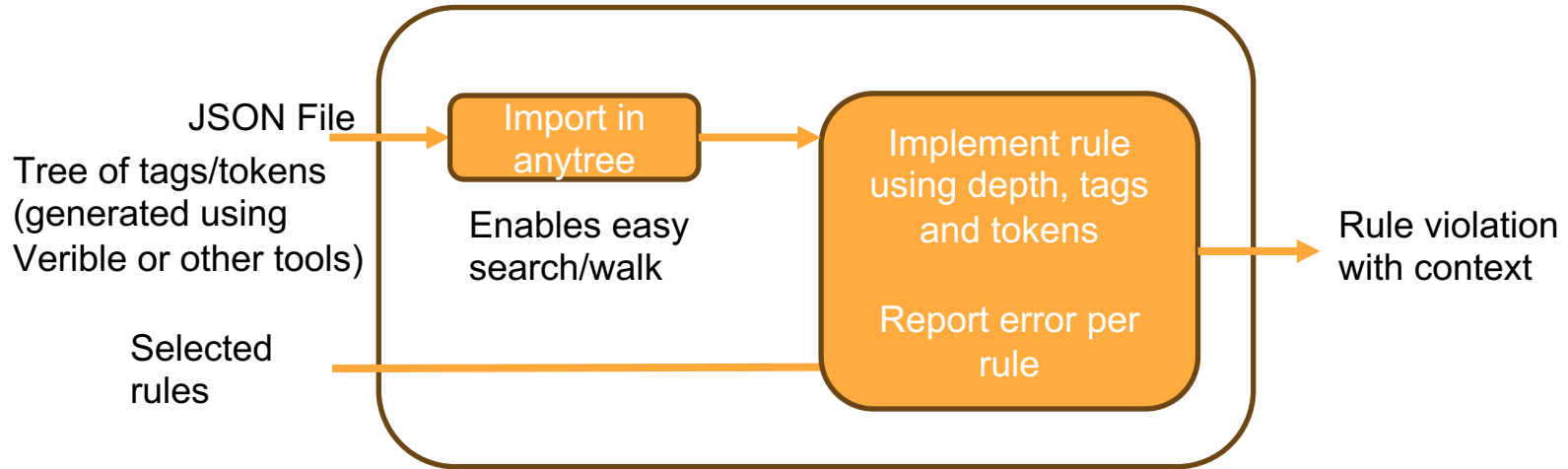
- Find bugs early
- Save time, cost
- Keep code compatible across tool chains

● Linters needed for:

- SystemVerilog RTL
- SystemVerilog TB
- UVM code base
- **Assertions**
- Acceleration
- Emulation
- Formal Verification

SVALint architecture

- SW architecture of SVALint using JSON input and anytree library
- Each rule is a few lines of code, thanks to the SVALint infrastructure



SVALint: By the Numbers & Value Proposition

25 lint rules across 5 categories — each engineered to catch a specific class of verification risk.

FUNCTIONALITY

7

Preventing false positives & robust action blocks.

STYLE

5

Unified naming conventions across verification teams.

PERFORMANCE

6

Preventing state-space explosion in Formal.

DEBUG

4

Isolating failures rapidly via clear labeling.

COMPATIBILITY

3

Ensuring safe, cross-tool IEEE 1800 syntax.

SVALint Rules: Functionality & Performance (1/2)

FUNCTIONALITY 7 rules

FuncNOLAPInCoverProp

Flags non-overlapped implications in covers (prevents false positives).

FuncOLAPInCoverProp

Flags overlapped implications in covers (prevents false positives).

NoWithinOperInAsrt

Avoids the misleading within sequence operator.

UseRealTimeVsTime

Recommends \$realtime instead of \$time.

FuncMissingFABLK

Checks for missing fail/action blocks in assertions.

PerfNoPABlk

Action block structural and performance checks.

ReuseNoOneLinerFABLK

Enforces reusable action block structuring.

PERFORMANCE 6 rules

NoRangeInAntAsrt

Avoids ranges in antecedents to prevent redundant thread generation.

NoUBRangeInAntAsrt

Restricts unbounded ranges (e.g., ##[1:\$]) to prevent state explosion.

NoUBRangeInConseqAsrt

Flags infinite ranges in consequents (they can never definitively fail).

NoFirstMatchOperAsrt

Discourages first_match() in formal to prevent state space explosion.

MissingImplicationOper

Flags SVA properties that omit implication operators.

NoLargeDelayProp

Flags SVA properties containing excessively large cycle delays.

SVALint Rules: Compatibility, Debug & Style (2/2)

COMPATIBILITY 3 rules

AvoidAAExistsSVA

Avoids `array.exists()` due to limited IEEE 1800 tool adoption.

AvoidPopBkSVA

Avoids `queue.pop_back()` due to limited tool adoption.

AvoidPopFrSVA

Avoids `queue.pop_front()` due to limited tool adoption.

DEBUG 4 rules

MissingLabelChk

Requires labels on assertions to enhance debuggability in waveforms.

MissingEndLblProp

Flags missing end-labels in property declarations.

MissingEndLblSEQ

Flags missing end-labels in sequence declarations.

UseSimpleExprConseq

Promotes simple consequents to easily isolate failures.

STYLE 5 rules

AssertNaming

Enforces `a_` prefix for assertions.

AssumeNaming

Enforces `m_` prefix for assumptions.

CoverNaming

Enforces `c_` prefix for cover statements.

PropNaming

Enforces standard naming conventions for property definitions.

NoExplTimeLiterals

Flags explicit time literals embedded directly within SVA code.

Rule example: Assert Prefix Rule

Code

```
// Assertion using the property missing  
a_prefix for the name  
  counter_limit_check: assert property  
(p_counter_within_limit)  
  else $error("Assertion Failed:  
Counter exceeds the maximum  
allowed value!");
```

```
// Assertion using the property with a  
prefix for the name  
  a_counter_limit_check: assert  
property (p_counter_within_limit)  
  else $error("Assertion Failed:  
Counter exceeds the maximum  
allowed value!");
```

tool output

```
python ../../bin/svalint.py -t test_2_tb_f.sv  
AsFigo: INFO: Loaded test file:  
test_2_tb_f.sv  
...  
AsFigo: Violation:  
[ASSERT_START_WITH_A]:  
Debug: Found assert name without a_  
prefix. Use a_ as assert prefix  
Severely impacts verification  
completeness as errors may go undetected  
counter_limit_check
```

```
python ../../bin/svalint.py -t test_2_tb_p.sv  
AsFigo: INFO: Loaded test file: test_2_tb_p.sv  
** Report counts by severity  
INFO : 12  
WARNING : 0  
ERROR : 0
```

```
-----  
AsFigo SVALint Report  
Summary  
Total lint rules executed: 10  
-----  
** Report counts by severity  
INFO : 12  
WARNING : 0  
ERROR : 1  
** Report counts by ID  
[ASSERT_START_WITH_A] 1
```


Summary

- Open-source linters allow adding your own rule
- SVALint flow
- Available rules (25)

Thanks!

SVALint – a BYOL: SystemVerilog Assertion Linter

For suggestions and feedback, email support@asfigo.com